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CE205
Databases and Information Retrieval

Anomalies, Inference Rules of Functional Dependencies, Keys and Attribute Closure

Dr. Zulfiqar Ali

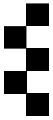
Lecturer

School of Computer Science and Electronic Engineering

Most of the slides based on

- *Database Systems: A Practical Approach to Design, Implementation, and Management* by Conoly and Begg, Global Edition/6th Edition.
- *Database Systems: Introduction to Databases to Data Warehouses* by Jukic, Vrbsky, Nestorov, sharma, 2nd Edition.

Functional Dependencies



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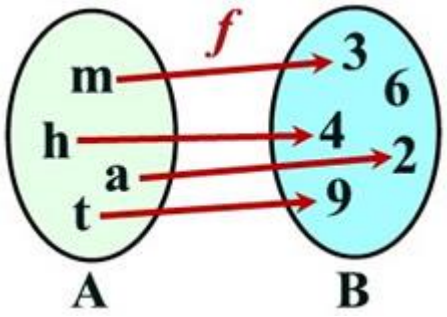
Functional Dependencies

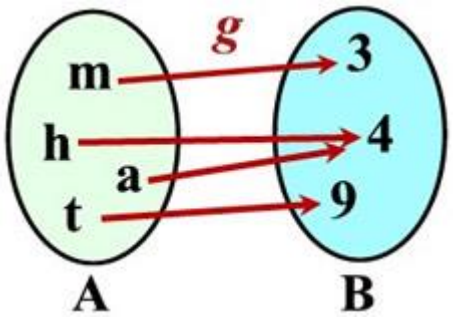
- **Functional dependency describes relationship between attributes.**
- **Important concept associated with normalization.**
  - 2NF, 3NF, BCNF
    - based on keys and FDs of a relation schema
  - 4NF
    - based on keys, multi-valued dependencies
  - 5NF
    - based on keys, join dependencies

Do You remember the Concept of Function studied in high school?

Function:

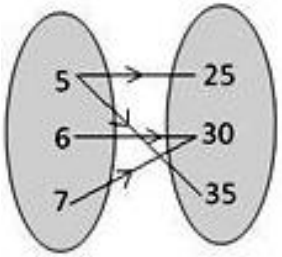
func will be function from *A* to *B*, *func:A*→*B*, if each element of *A* assigned to a unique element of **B**. For example, f and g are both functions.



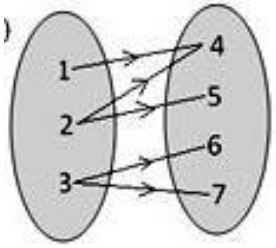


Identify Functions

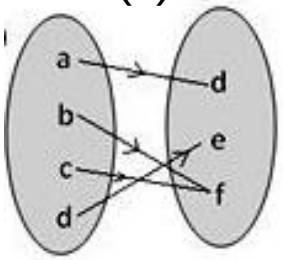
(a)



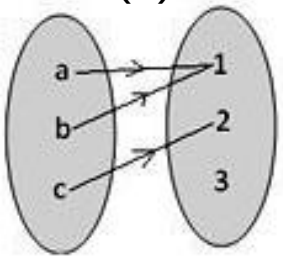
(b)



(c)



(d)



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Functional Dependencies (FD)

- A functional dependency occurs when the value of one (set of) attribute(s) uniquely determines the value of a second (set of) attribute(s) in the same record of the relation..

ObjectColor → Weight

ObjectColor → Shape

ObjectColor	Weight	Shape
Red	5	Ball
Blue	5	Cube
Yellow	7	Cube

- The attribute on the left side of the functional dependency is called the determinant.

ObjectColor	Weight	Shape
Red	5	Ball
Blue	5	Cube
Red	7	Cube

Is valid now?

ObjectColor → Weight

ObjectColor → Shape

NO!

Equations and Functional Dependencies

- **Functional dependencies may be based on equations:**

  ExtendedPrice = Quantity X UnitPrice

  (Quantity, UnitPrice) → ExtendedPrice

  Functional dependencies are not equations!

- Other notation for functional dependency:



(Graphical representation of FDs)

What FDs may exist?

Consider a sample relation *R*(A, B, C, D).Which FDs *may exist* in this relation?

Answer:

Some FD which may exist are

B→C
{A,B}→C
{C,D}→D

Why A→B and B→D are not valid?

A	B	C	D
a1	b1	c1	d1
a1	b2	c2	d2
a2	b2	c2	d3
a3	b3	c4	d3

FDs in Relation AD CAMPAIGN MIX

AD CAMPAIGN MIX

AdCampaignID	AdCampaign Name	StartDate	Duration	Campaign MgrID	Campaign MgrName	ModusID	Media	Range	Budget Pctg
111	SummerFun20	6.6.2020	12 days	CM100	Roberta	1	TV	Local	50%
111	SummerFun20	6.6.2020	12 days	CM100	Roberta	2	TV	National	50%
222	SummerZing20	6.8.2020	30 days	CM101	Sue	1	TV	Local	60%
222	SummerZing20	6.8.2020	30 days	CM101	Sue	3	Radio	Local	30%
222	SummerZing20	6.8.2020	30 days	CM101	Sue	5	Print	Local	10%
333	FallBall20	6.9.2020	12 days	CM102	John	3	Radio	Local	80%
333	FallBall20	6.9.2020	12 days	CM102	John	4	Radio	National	20%
444	AutmnStyle20	6.9.2020	5 days	CM103	Nancy	6	Print	National	100%
555	AutmnColors20	6.9.2020	3 days	CM100	Roberta	3	Radio	Local	100%

Determine some of FDs in the relation

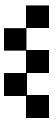
AdCampaignID→AdCampaignName

AdCampaignID→StartDate

AdCampaignID→Duration

AdCampaignID→CampaignMgrID

AdCampaignID→CampaignMgrName



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Functional Dependency Rules

Decomposition/Splitting rule

- **If A → (B, C), then A → B and A→ C.**

If

EmployeeID → {Name, Address}

Then

EmployeeID → Name and EmployeeID → Address

Union/Combining rule is reverse of the splitting rule.

- **If A → B and A → C, then A → (B, C).**

Left hand side of a FD cannot be split and combined.

Combine FDs of Relation AD CAMPAIGN MIX

AD CAMPAIGN MIX

AdCampaignID	AdCampaign Name	StartDate	Duration	Campaign MgrID	Campaign MgrName	ModusID	Media	Range	Budget Pctg
111	SummerFun20	6.6.2020	12 days	CM100	Roberta	1	TV	Local	50%
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555	AutmnColors20	6.9.2020	3 days	CM100	Roberta	3	Radio	Local	100%

Determine some of FDs in the relation

AdCampaignID→AdCampaignName

AdCampaignID→StartDate

AdCampaignID→Duration

AdCampaignID→CampaignMgrID

AdCampaignID→CampaignMgrName

After combing, we get set1:

AdCampaignID→AdCampaignName, StartDate, Duration, CampaignMgrID, CampaignMgrName

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Inference Rules
Armstrong's Axioms for FDs

This is the syntactic way of computing/testing the various properties of FDs

- **Reflexivity:** If $Y \subseteq X$ then $X \rightarrow Y$ (also known as trivial FD)
  - Name, Address $\rightarrow$ Name or Name, Address $\rightarrow$ Address
- **Augmentation:** If $X \rightarrow Y$ then $X Z \rightarrow YZ$
  - If Town $\rightarrow$ Zip
    then Town, *Name* $\rightarrow$ Zip, *Name*
- **Transitivity:** If $X \rightarrow Y$ and $Y \rightarrow Z$ then $X \rightarrow Z$
  - If EmployeeID $\rightarrow$ Town and Town $\rightarrow$ Zip,
    then EmployeeID $\rightarrow$ Zip

Equivalent FDs

- Equivalent functional dependency - occurs when two columns (or sets of columns) that functionally determine each other determine other columns
- If one of the equivalent functional dependencies is depicted, the other equivalent functional dependency can be omitted
- For example: if functional dependencies :
  $A \rightarrow B$ and $B \rightarrow A$ exists in a relation, then :

$A \rightarrow B$ and $B \rightarrow A$ are equivalent	$A \rightarrow \{B, X\}$ and $B \rightarrow \{A, X\}$ are equivalent	$\{Y,A\} \rightarrow \{B, X\}$ and $\{Y,B\} \rightarrow \{A, X\}$ are equivalent

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All FD in Relation AD CAMPAIGN MIX (1)

AD CAMPAIGN MIX

AdCampaignID	AdCampaign Name	StartDate	Duration	Campaign MgrID	Campaign MgrName	ModusID	Media	Range	Budget Pctg
111	SummerFun20	6.6.2020	12 days	CM100	Roberta	1	TV	Local	50%
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555	AutmnColors20	6.9.2020	3 days	CM100	Roberta	3	Radio	Local	100%

Determine all FDs in the relation

set1:

AdCampaignID→AdCampaignName, StartDate, Duration,
CampaignMgrID, CampaignMgrName

All FD in Relation AD CAMPAIGN MI) (2)

Initially recognized sets of functional dependencies in the relation AD CAMPAIGN MIX

(Set 1) AdCampaignID → AdCampaignName, StartDate, Duration, CampaignMgrID, CampaignMgrName

(Set 2) AdCampaignName → AdCampaignID, StartDate, Duration, CampaignMgrID, CampaignMgrName

(Set 3) CampaignMgrID → CampaignMgrName

(Set 4) ModusID → Media, Range

(Set 5) AdCampaignID, ModusID → AdCampaignName, StartDate, Duration, CampaignMgrID, CampaignMgrName, Media, Range, BudgetPctg

(Set 6) AdCampaignName, ModusID → AdCampaignID, StartDate, Duration, CampaignMgrID, CampaignMgrName, Media, Range, BudgetPctg

All FD in Relation AD CAMPAIGN MIX (3)

Initially recognized sets of functional dependencies in the relation AD CAMPAIGN MIX

(Set 1) AdCampaignID → AdCampaignName, StartDate, Duration, CampaignMgrID, CampaignMgrName

(Set 3) CampaignMgrID → CampaignMgrName

(Set 4) ModusID → Media, Range

(Set 5) AdCampaignID, ModusID → BudgetPctg

Final Remarks

Now you Know

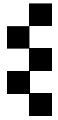
- Update Operations
- Why redundant data is not good in a relation
- What are update anomalies
- How to find FDs in a relation

For normalisation of a relation, still we need to know

- How to find Candidate key(s) in a relation
- What are partial FD and Transitive FD
- Dependency Preserving Decomposition
- Lossless Join Decomposition

Final Destination

2NF, 3NF, BCNF, some info about (4NF, 5NF)



Suggested Readings

- D1: Database processing: fundamentals, design, and implementation (David M. Kroenke; David J. Auer) . 14th Edition (Global Edition). Pearson.
  - Chapter 3
- D2: The following sections from the textbook *Database Systems: A Practical Approach to Design, Implementation, and Management* by Conoly and Begg, Global Edition/6th Edition.
  - 14.2, 14.3, 14.4, 15.1
- https://opentextbc.ca/dbdesign01/chapter/chapter-11-functional-dependencies/